ScAN - A Path to PIM Hardware-aware Network

Scope

Tracks CIFAR-100 from RGB/FP32 references to RAW-RGGB, 4W4A, QKD, and the first PIM-conv simulator path. Unless stated otherwise, quantized accuracy uses the fixed simulator with raw_noise=True on 50k train / 10k test.

Pipeline Summary

Stage	Input	Model	Inference path	Current status
RGB FP32	RGB CIFAR-100	legacy FP32 CNN	floating point	Legacy reference only; rerun not yet in this repo.
RAW/RGGB FP32	4-channel RGGB	ALL-CNN-C + BN	floating point	Current reproducible baseline.
RAW/RGGB W4A4	4-channel RGGB	ALL-CNN-C QAT	INT simulator	Uniform W4A4 baseline validated.
QKD uniform	4-channel RGGB	QAT student + EfficientNetV2-L teacher	INT simulator	Matched-schedule partial rerun was below non-uniform path.
QKD non-uniform	4-channel RGGB	QAT + integer LUT-Q + PACT	codebook INT simulator	Current best validated dense int4 checkpoint.
PIM conv	4-channel RGGB	PIM-QAT student	PIM simulator	Simulator path implemented; PIM-specific training is next.

Student Network

Layer	Operation	Cin -> Cout	Kernel	Stride	Output
1.1	Conv+BN+ReLU	4 -> 96	3x3	1	96 x 16 x 16
1.2	Conv+BN+ReLU	96 -> 96	3x3	1	96 x 16 x 16
1.3	Conv+BN+ReLU	96 -> 96	3x3	2	96 x 8 x 8
2.1	Conv+BN+ReLU	96 -> 192	3x3	1	192 x 8 x 8
2.2	Conv+BN+ReLU	192 -> 192	3x3	1	192 x 8 x 8
2.3	Conv+BN+ReLU	192 -> 192	3x3	2	192 x 4 x 4
3	Conv+BN+ReLU	192 -> 192	3x3	1	192 x 4 x 4
4	Conv+BN+ReLU	192 -> 192	1x1	1	192 x 4 x 4
5	Conv logits + GAP	192 -> 100	1x1	1	100 x 4 x 4 -> 100

Current Results

Experiment	Precision	Eval path	Acc.	Notes
RGB FP32 reference	FP32	float	pending	Legacy RGB result should be rerun if needed for strict comparison.
RAW/RGGB FP32	FP32	float	61.31%	Default noisy test sanity check; training-log best 61.44.
RAW/RGGB FP32	FP32	float	61.18%	Clean test, raw_noise=False.
RAW/RGGB QAT	W4A4 uniform	INT simulator	56.61%	Validated noisy simulator baseline.
RAW/RGGB QAT	W4A4 uniform	INT simulator	57.09%	Clean simulator reference from previous run.
QKD uniform, first pass	W4A4 uniform	INT simulator	56.02%	Worse than QAT; not accepted as final QKD.
QKD uniform, matched schedule	W4A4 uniform	INT simulator	56.93%	Partial CS rerun best; below non-uniform path.
QKD non-uniform SS	W4A4 LUT-Q, m=7	codebook INT	56.78%	Integer LUT values [-7,...,7].
QKD non-uniform SS	W4A4 LUT-Q, m=15	codebook INT	56.52%	Integer LUT values [-14,-12,...,14]; training-log best 57.06.
QKD non-uniform SS + PACT	I16/W4/A4 LUT-Q, m=15	codebook INT	60.21%	Full rerun; training-log best 60.66.
QKD non-uniform CS + PACT	I16/W4/A4 LUT-Q, m=15	codebook INT	61.24%	Full rerun; training-log best 61.39.
QKD non-uniform TU + PACT	I16/W4/A4 LUT-Q, m=15	codebook INT	61.66%	Full rerun; training-log best 61.81.

QKD Setup

Teacher checkpoint b4_100.pth uses 4-channel RAW/RGGB input, not RGB. Teacher preprocessing upsamples RGGB tensors to 224 x 224 and normalizes with (x-0.5)/0.5.

Teacher / stage	Acc.	Notes
EfficientNetV2-L teacher, clean	77.30%	Verified on current H5 test split.
EfficientNetV2-L teacher, noisy	77.33%	Default noisy evaluation.
Co-studied teacher, noisy	78.70%	Earlier short CS run improved teacher.
QKD SS rerun	60.21%	I16/W4/A4 LUT-Q + PACT, fixed simulator.

Teacher / stage	Acc.	Notes
QKD CS rerun	61.24%	I16/W4/A4 LUT-Q + PACT, fixed simulator.
QKD TU rerun	61.66%	Current best dense int4 simulator result.

Integer Simulator

Dense int4 simulator: unsigned I16 first-layer input [0,65535], unsigned A4 hidden activations [0,15], uniform W4 [-7,7] or non-uniform LUT-Q indices, int32 accumulation, BN folded before execution, and precomputed requantization. Runtime forward does not run tanh, k-means, nearest-LUT search, or weight quantization.

Setting	Fake QAT	INT simulator
Noisy, raw_noise=True, num_workers=0	56.61%	56.61%
Clean, raw_noise=False, num_workers=0	57.09%	57.09%
Non-uniform SS, integer LUT-Q m=7, noisy	56.78%	56.78%
Non-uniform SS, integer LUT-Q m=15, noisy	56.52%	56.52%
Non-uniform SS + PACT, integer LUT-Q m=15, I16 input, noisy	--	60.21%
Non-uniform CS + PACT, integer LUT-Q m=15, I16 input, noisy	--	61.24%
Non-uniform TU + PACT, integer LUT-Q m=15, I16 input, noisy	--	61.66%

PIM Conv Status

Implemented explicit PIM-QAT model and PIM simulator paths without changing the default dense-QKD path. Current minimal PIM path keeps the existing BN student model; GBN/PCN are not enabled in this pass.

PIM path follows pim_info/scanpcn_inference_spec.tex: row select over [18,18,36,72], keep 48/144 by default, hybrid stage1.1 keep 72/144, 7-bit mid-rise tile ADC, digital BN scale/bias, normalized hidden DAC activations in [0,1], and unmasked stage1.0 / logits layers. PIM-QAT eval and PIM simulator match on TU checkpoint smoke input: max_abs_diff <= 6e-7.

Current Best

checkpoints/rerun/qkd_i16_intlut_pact_g8m15_lutq_tu.pth

Next Updates

Train the PIM-QAT model from the current dense int4 checkpoint, report all new PIM results through pim_sim_test with raw_noise=True, and add GBN/PCN only after the minimal BN PIM path is stable.